Physical Chemistry CUME, Nov. 18, 2016

Quantum Chemistry

The quantum state of a particle is described by time-independent wavefunction $\psi_0(x)$ defined along the x axis:

$$\psi_0(x) = A \exp \frac{-x^2}{2\sigma^2},$$

where σ is a positive real number and $A = |A| \exp(i\varphi)$ is generally a complex number, whose absolute value and phase are denoted by $|A|$ and φ, respectively.

Q1. If $|\psi_0(x)|^2 dx$ is the probability of finding the particle between x and $x + dx$ at time $t = 0$, determine the absolute value and phase of A?

Q2. The particle described above is in a harmonic potential, i.e., $V(x) = \frac{1}{2}kx^2$, where $k = \hbar^2/(m\sigma^4)$ is the harmonic constant and m is the particle mass. Is $\psi_0(x)$ an eigenfunction of the Hamiltonian ($\widehat{H}$)? Provide proof of your answer.

Q3. For the particle described above, find the expectation value of the total energy (Hamiltonian) in terms of reduced Planck constant $\hbar$ and angular frequency $\omega \equiv \sqrt{k/m}$.

Q4. For the particle described above, find the expectation value of the potential energy in terms of $\hbar$ and ω.

Q5. The particle described above absorbs a photon of wavelength $2\pi c/\omega$, where c is the speed of light and ω is defined in **Q3**. For a particle at this excited state, find the expectation value of the Hamiltonian?

Note: Each Problem has 20 points.

Useful Relations:

$$\widehat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)$$

$$\int_{-\infty}^{\infty} \exp\frac{-x^2}{\sigma^2} dx = \sqrt{\pi}\sigma, \qquad \int_{-\infty}^{\infty} x^2 \exp\frac{-x^2}{\sigma^2} dx = \frac{1}{2}\sqrt{\pi}\sigma^3$$

$$\frac{\partial^2}{\partial x^2} \exp\frac{-x^2}{2\sigma^2} = \left(\frac{-1}{\sigma^2} + \frac{x^2}{\sigma^4}\right) \exp\frac{-x^2}{2\sigma^2}$$